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2N5133

- LOW NOISE - $NF = 1.5 \text{ dB (TYP) @ } 1.0 \text{ kHz}$
- HIGH GAIN - $h_{FE} = 60 \text{ (MIN), } 220 \text{ (TYP) @ } 1.0 \text{ mA}$
 $h_{FE} = 50 \text{ (TYP) @ } 50 \mu\text{A}$
- BREAKDOWN VOLTAGE - $V_{CEO} = 18 \text{ VOLTS (MIN)}$

ABSOLUTE MAXIMUM RATINGS (Note 1)

Maximum Temperatures

Operating Junction Temperature

Storage Temperature

Lead Temperature (Soldering, 10 second time limit)

125°C Maximum
-55°C to +125°C
260°C Maximum

Maximum Power Dissipation

Total Dissipation at 25°C Case Temperature (Note 2)

at 25°C Ambient Temperature (Note 2)

0.5 Watt
0.2 Watt

Maximum Voltages and Current

V_{CBO} Collector to Base Voltage

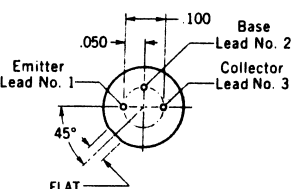
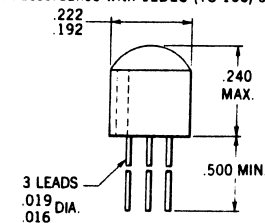
V_{CEO} Collector to Emitter Voltage (Note 3)

V_{EBO} Emitter to Base Voltage

20 Volts
18 Volts
3.0 Volts

PHYSICAL DIMENSIONS

In accordance with JEDEC (TO-106) outline



NOTES: All dimensions in inches
All leads electrically isolated from case
Package weight is 0.31 gram Package
is electrically non-conductive material

ELECTRICAL CHARACTERISTICS (25°C Free Air Temperature unless otherwise noted)

SYMBOL	CHARACTERISTICS	MIN.	TYP.	MAX.	UNITS	TEST CONDITIONS
h_{FE}	DC Current Gain	60	220	1000		$I_C = 1.0 \text{ mA}$ $V_{CE} = 5.0 \text{ V}$
h_{FE}	DC Current Gain		50			$I_C = 50 \mu\text{A}$ $V_{CE} = 10 \text{ V}$
h_{fe}	High Frequency Current Gain ($f = 20 \text{ MHz}$)		1.3			$I_C = 50 \mu\text{A}$ $V_{CE} = 5.0 \text{ V}$
h_{fe}	High Frequency Current Gain ($f = 20 \text{ MHz}$)	2.0		20		$I_C = 1.0 \text{ mA}$ $V_{CE} = 5.0 \text{ V}$
NF	Narrow Band Noise Figure ($f = 1.0 \text{ kHz}$)		1.5		dB	$I_C = 30 \mu\text{A}$ $V_{CE} = 5.0 \text{ V}$ $PWR \text{ BW} = 200 \text{ Hz}$ $R_S = 10 \text{ k}\Omega$
$V_{CE(sat)}$	Collector Saturation Voltage			0.4	Volts	$I_C = 1.0 \text{ mA}$ $I_B = 0.1 \text{ mA}$
I_{CBO}	Collector Cutoff Current			50	nA	$I_E = 0$ $V_{CB} = 15 \text{ V}$
$I_{CBO}(65^\circ\text{C})$	Collector Cutoff Current			5.0	μA	$I_E = 0$ $V_{CB} = 15 \text{ V}$
C_{cb}	Collector-Base Capacitance			5.0	pF	$I_E = 0$ $V_{CB} = 5.0 \text{ V}$
BV_{CBO}	Collector to Base Breakdown Voltage	20			Volts	$I_C = 100 \mu\text{A}$ $I_E = 0$
$V_{CEO(sust)}$	Collector to Emitter Sustaining Voltage (Notes 3 and 4)	18			Volts	$I_C = 3.0 \text{ mA}$ $I_B = 0$
BV_{EBO}	Emitter to Base Breakdown Voltage	3.0			Volts	$I_E = 10 \mu\text{A}$ $I_C = 0$
$V_{BE(on)}$	Base to Emitter On Voltage			0.75	Volts	$I_C = 100 \mu\text{A}$ $V_{CE} = 5.0 \text{ V}$
h_{fe}	Small Signal Current Gain ($f = 1.0 \text{ kHz}$)	50		1100		$I_C = 1.0 \text{ mA}$ $V_{CE} = 5.0 \text{ V}$

*Planar is a patented Fairchild process.

NOTES:

- (1) These ratings are limiting values above which the serviceability of any individual semiconductor device may be impaired.
- (2) These ratings give a maximum junction temperature of 125°C and junction to case thermal resistance of 200°C/Watt (derating factor of 5.0 mW/°C); junction to ambient thermal resistance of 500°C/Watt (derating factor of 2.0 mW/°C).
- (3) Rating refers to a high-current point where collector to emitter voltage is lowest. For more information send for Fairchild Publication APP-4/2.
- (4) Pulse Conditions: length = 300 μs ; duty cycle = 1%.